

Data Structures & Algorithms 2026

Final Projects

1 Instructions

1. Your goal is to create BOTH:

A report (typeset PDF) on the background and theory.

Code repository with source, tests, README, and runners to demo part of the theory.

2. You will be in teams of 6 (± 1 for any remainders).

I will give a short starting point for the topic, and you may consult for general guidance, but there are **no additional prompts**.

You must go to sufficient technical and theoretical depth under your own power.

3. You may propose your own topics, but I also have a list of suggested topics below.

Proposed topics must have similar theoretical depth in order to be approved.

No repeated / old topics between teams (first-come-first-served).

4. Do not spend any money on this project.

5. There will be a session in the middle of the term where the project proposals should be finalized, and prototype code / notebooks submitted.

Final submissions are due at the end of the term.

The final session will also include 10-minute presentations from each team (subject to approval).

2 Rubric

Theory Report [40%]

- Theoretical Depth and Completeness - 40%
- Conciseness & Elegance of Prose - 40%
- Grammar and Professionalism - 20%

Code Repository [40%]

- Code Performance / Implementation Success - 40%
- Complete Automated Testing Suite (unit, integration, stress, etc.) - 40%
- Code Hygiene, Organization, Readability / PEP8 - 20%

Presentation [20%]

- Visualizations and Slides - 40%
- Completeness and Clarity - 40%
- Timeliness & Professionalism - 20%

3 Suggested Topics

1. **Quantum Algorithms I : Quantum Speedup** (Linear Algebra, Analysis of Algorithms, DMW)
 - Shor's Algorithm
 - Grover's Algorithm
 - HHL Algorithm
 - Kerenidis & Prakash - Least-Squares, Recommender Systems
2. **Quantum Algorithms II : De-quantized Models** (Linear Algebra, Analysis of Algorithms, DMW)
 - Recommender Systems
 - Principal Component Analysis
 - Linear Regression
3. **Orderbook Construction** (Sorting, Time Complexity, Locality, FPGA Engineering)
4. **Competitive Algorithms: Catch-up in Guess Who** (Sorting, DP)
5. **Competitive Algorithms: Optimal Territory War** (Analysis of Algorithms)
6. **Pokemon Safari Optimization** (DP)
 - See also: Catching-up Problem vs rocks
7. **Thoughtful Solitaire** (DP, Sorting, Heuristics)
8. **Chinese Poker** (Sorting, DP, Game Theory, Counterfactual Regret Minimization)
9. **MOSS and Plagiarism Detectors** (Hashing, Information Retrieval)
10. **Fast Fourier Transforms** (Linear Algebra, Discrete Math, Signal Analysis)
 - See also: Autotune, Spectrograms, MUSIC
11. **Hashing Extended - Bloom Filters, LSH, Proof-of-work / Bitcoin** (Hashing)
12. **Cryptographic Hashing for Zero-knowledge Proof** (Hashing, Low-latency ML Use)
13. **Spellsided Optimizer** (Graph Algorithms, Heuristics, Combinatorial Optimization)
14. **CPython Algorithms (Deep Dive)** (Miscellaneous, Python, C)
15. **Nonograms** (Heuristics, 2-SAT)
16. **TeXnique** (LaTeX math, Abstract Syntax Trees, Structural Equality Checking, pyLatex)
17. **Foraging Theory - Optimal, Suboptimal, Bayesian, etc.** (DP, LP, Ecology)
18. **Factorio: *The Factory Must Grow*** (Operations Research / Multiple)
 - Heuristic Optimization (Simulated Annealing, Genetic Algorithms, Particle Swarm)
 - Conveyor Belts & Automated Sortation Systems
 - Fluids & Pipe Networks (DP)
 - Vehicle Routing & Deadlocks

19. **Tetris** (Complexity Classes, Sorting)
20. **Magic: The Gathering as Programming** (Turing Completeness, Automata)
Alternatively: Powerpoint for Programming
21. **Compression Algorithms - Lossless and Lossy** (Data Structures, Math)
See also: Huffman, Lempel-Ziv, PPM, vs LLMs as a natural language compressor.
22. **Project Management Algorithms** (Automata, Heuristics)
See also: Uber
23. **Sokoban / Warehouse Optimization** (Complexity Classes, Heuristics / Pruning)
Recommended introductory reference: Timo Virkkala
24. **The Stepping Stones Puzzle** (Combinatorial Optimization)
25. **Pebbling Chessboards** (Graphs, Complexity Classes)
26. **Rubik's Cube Algorithms, God's Algorithm** (Graphs, Group Theory)
27. **Netcode, Rollback Algorithms, Buffers, Latency, GGPO** (Data Structures, Distributed Systems, Version Control)
28. **Knapsack Problem - Regular, Robust, Robust Polynomial, Cryptographic**
(Complexity Classes, Heuristics)
29. **Bakeries** (Concurrency, Data Structures, GIL)
30. **Minesweeper** (Complexity Classes)
31. **Origami / Computational Folding** (Geometry, Math, Complexity Classes)
32. **Algorithmic Music Composition** (Heuristics, Signal Analysis, DAGs)
33. **Chess / Shogi Engines** (Search, Evaluation, DFS, Heuristics)
34. **Bioinformatics** (Various)
Gene Regulatory Networks
DNA $\rightarrow$ Protein Translation as Source $\rightarrow$ Compiled Code
fold.it
35. **TAS:Tool-assisted Speedruns, Pseudo-Randomness** (DP)
36. **Genetic Algorithms** (Heuristics - Genetic Algorithms)
37. **Ising Machines** (Complexity Classes, Heuristics - Simulated Annealing)
38. **Computing, Proof-Assists, Specification Language** (Proofs, Formal Logic)
See also: computer-assisted proofs, 4-color theorem